Supination	Lateral rotation of the radius, resulting in the palm of the hand facing anteriorly (if in anatomical position) or superiorly (if elbow is flexed). (**Mnemonic: Supinate to the Sun and Pronate to the Plants.** Supinate: palm towards the Sun; Pronate: palm towards the Plants)
Pronation	Medial rotation of the radius, resulting in the palm of the hand facing posteriorly (if in anatomical position) or inferiorly (if elbow is flexed)
Protrusion	Forward
Retrusion	Backward
Elevation	Superiorly to the reference axis
Depression	Inferiorly to the reference axis
Circumduction	Combination of flexion, abduction, extension, and adduction
Deviation	Ulnar and radial abduction of the wrist
Opposition	Bringing the thumb in contact to a finger
Reposition	Separating the thumb from the digits
Plantar Flexion	The movement of the foot in a downward motion away from the body
Dorsal Flexion	The movement that occurs when the foot and shin approximate closer together

1.2. CELLULAR LEVEL OF ORGANIZATION

❑ Introduction to the Cell

All living things are composed of cells. Molecules combine to form cells. A cell is defined as the structural and functional unit of the living body. It can live, grow, and reproduce on its own. Cells are often called the building blocks of life. Each type of cell is made of chemicals and carries out specific chemical reactions.

❑ History of Cell

- **Robert Hooke** discovered the cell in **1665.** He gave the name of his discovery "Cells", which means "a small room" in Latin.
- Anton Van Leeuwenhoek first saw and described a live cell.
- In 1883, Robert Brown later discovered the Nucleus.

❑ Cell Theory

- Cell theory was jointly put forward by **Matthias Jacob Schleiden** and **Theodor Schwann** in 1839.
- Rudolf Virchow (1855) first explained that cells divide and new cells are formed from preexisting cells.

❑ Types of Cells

- Cells are broadly divided into two major types:
  - ✓ **Prokaryotic cells**
  - ✓ **Eukaryotic cells**

Table 1.6: Comparison of Prokaryotic and Eukaryotic Cells

CHARACTERISTIC	PROKARYOTIC CELLS	EUKARYOTIC CELLS
Complexity	Simple in structure.	More complex in structure.
Size	Small (generally **1–5 µm**).	Large (generally **5–100 µm**).
Organization	Usually unicellular.	May be unicellular or multicellular.
Cell Covering	Cell is enclosed by a rigid **cell wall**.	Cell is enclosed by a flexible **plasma membrane**.
Nucleus	Lack a true membrane-bound nucleus.	Possess a well-defined nucleus.
Subcellular Organelles	Membrane-bound organelles are absent.	Distinct membrane-bound organelles are present (e.g., mitochondria, nucleus, lysosomes).

Unlock FREE B.Pharm 1st Semester Question Practice & e-PDFs - Download GDC CLASSES App Now

Genetic Material	DNA is located in the cytoplasm within the nucleoid region.	DNA is enclosed within the nucleus.
Cell Division	Usually occurs by **binary fission**; mitosis is absent.	Cell division occurs by **mitosis** (and meiosis in reproductive cells).
Cytoplasm	Membrane-bound organelles and cytoskeleton are absent.	Contains membrane-bound organelles and a cytoskeleton (a network of tubules and filaments).
Ribosomes	**70S ribosomes** are present.	**80S ribosomes** are present.
Examples	Bacteria and Archaea.	Animals, plants, fungi, and protists.

1.2.1. Structure & Functions of The Cell:

- A cell consists of a **plasma membrane** enclosing a number of **organelles suspended in a watery fluid called cytosol.**
- The cell contents, excluding the nucleus, are made up of the cytoplasm, namely the **cytosol and other organelles.**

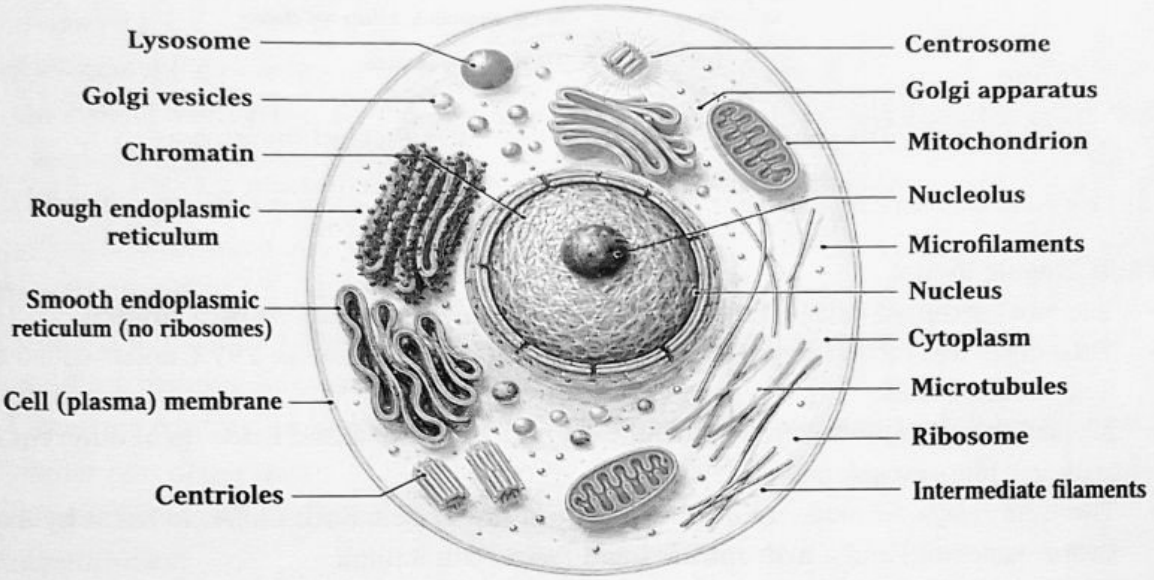


Fig.1.7: Structure of a Typical Human Cell

- Cell can be divided into three parts:
  - ✓ **Plasma Membrane**
  - ✓ **Cytoplasm**
  - ✓ **Nucleus**

❑ Plasma Membrane

- The plasma membrane also called **plasmalemma or cell membrane** is the outer covering of the cell.
- It is **selectively permeable,** which means it allows some substances to pass through while blocking others. It is flexible, responsive, and constantly changing in structure.
- This membrane separates one cell from another and helps maintain the internal environment of the cell by controlling the movement of ions and nutrients.
- It also has receptors that help in recognizing small molecules and plays an important role in interaction between cells.
- Under an electron microscope, the cell membrane appears about **7.2–8 nm thick.**
- It has a trilaminar (three-layered) structure, with proteins and sugars present within it.

- This three-layered structure is known as the **unit membrane**.
- It is made up of a **double layer (bimolecular layer) of lipid molecules**, which looks light under the microscope.
- This **lipid layer** is present in the middle and is surrounded on both sides by **two darker protein layers**.

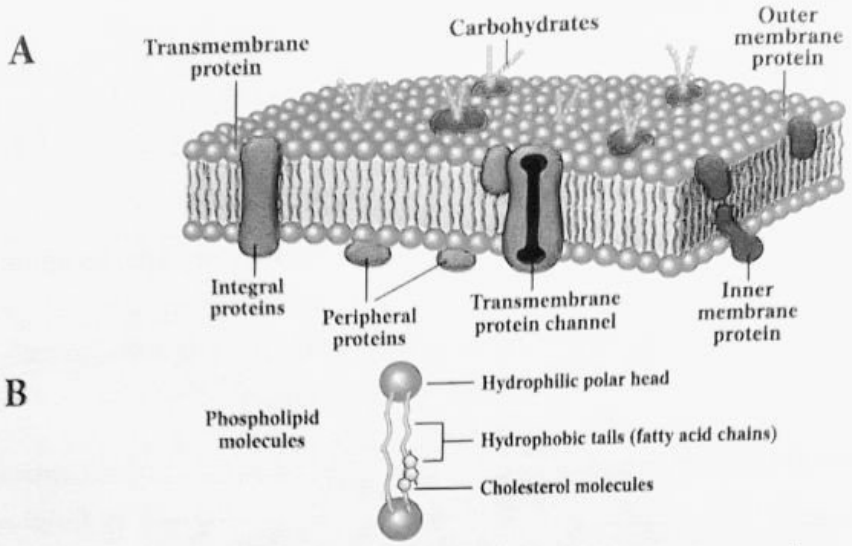


Fig.1.8: Diagrammatic representation of a Plasma membrane
A. Fluid Mosaic Model of the Plasma Membrane
B. Structure of a Phospholipid Molecule

❖ **Fluid Mosaic Model**

- The most accepted model of the cell membrane today is the **fluid mosaic model**.
- This model was further improved by **SJ Singer and GL Nicolson in 1972**, and is called the modified fluid mosaic model.
- According to this model, the cell membrane is flexible (fluid) and made up of different components arranged like a mosaic pattern.
- The lipid bilayer is made up of **phospholipid molecules**. Each molecule has a **hydrophobic tail** (water repelling) and a **hydrophilic head** (water-attracting).
- The hydrophilic heads face the outer and inner sides of the membrane, while the hydrophobic tails face towards the inside of the membrane.
- This arrangement helps to keep the inside and outside of the cell separate.

❖ **Protein Layers of Cell Membrane**

- The protein layers of the cell membrane are dense layers that cover both sides of the central lipid layer. These protein layers **protect the lipid layer.** The substances present in these layers are mainly glycoproteins.
- Membrane proteins run across the cell membrane and create pathways that help electrolytes and nutrients move in and out of the cell. These proteins have different roles:
  - ✓ Some work like enzymes.
  - ✓ Some act as receptors that attach to specific hormones or other substances.
  - ✓ Some form channels for the movement of small, water-soluble ions.
  - ✓ Their carbohydrate part helps the cell to be recognized (immunological identity).
  - ✓ They also help in the active transport of substances across the membrane.
- The carbohydrates of the membrane are present as **glycoproteins and glycolipids**. These carbohydrate parts extend outward from the cell surface. Both glycoproteins and glycolipids function as receptor sites.

Unlock FREE B.Pharm 1st Semester Question Practice & e-PDFs - Download GDC CLASSES App Now

❑ Cytoplasm

- **Cytoplasm** is the part of the cell that surrounds the nucleus and is enclosed by the cell membrane. It **houses all organelles** (like mitochondria and ribosomes) except the nucleus, making it the site of metabolic activities such as protein synthesis.
- **Protoplasm comprises both the cytoplasm and the nucleus**. Cytoplasm is clear and gel-like in appearance. It mainly consists of water and contains organelles, enzymes, salts, and different organic substances. Organelles such as **ribosomes, mitochondria, endoplasmic reticulum, and vacuoles** are present in it.
- When observed under a light microscope, the cytoplasm is divided into two main parts:
  - ✓ **Cytoskeleton**
  - ✓ **Cytoplasmic organelles and inclusions**

❖ Cytoskeleton

- It is a **network of protein fibers**, that makes up the cytoskeleton.
- It supports the cell, keeps it in form, and aids in organelle organization and retention.
- It consists of:

➢ Intermediate Filaments:

- Intermediate filaments are important components of the cell cytoskeleton that **provide mechanical strength and protect the cell from stress.**
- They help **maintain the position and stability** of organelles, especially the nucleus.
- They help form cell junctions and provide strength and support to tissues.

➢ Microfilaments:

- Microfilaments are **very fine fibers made of actin** that help in **cell movement, maintain cell shape,** and are involved in **muscle contraction.**
- They are **contractile proteins measuring about 4–6 nm in diameter** and are mainly located at the outer part of the cell.
- These fibers provide support and help the cell change shape when needed.

➢ Microtubules:

- Microtubules are **long, hollow, unbranched tubes** that act as tracks for the movement of mitochondria, chromosomes, and secretory granules within the cytoplasm.
- They are rigid structures that help **maintain cell shape and provide support to the cell.**

❖ Cytoplasmic Organelles

- Organelles mean **"small organs."** They have specific and specialized functions inside the cell and are usually surrounded by a membrane in the cytoplasm.
- Examples include the **nucleus, mitochondria, ribosomes, endoplasmic reticulum, Golgi apparatus, lysosomes, and cytoskeleton**.

➢ Endoplasmic Reticulum

- The endoplasmic reticulum (ER) is composed of a network of vesicles (cisternae) and tubes (tubules). It is a **three-dimensional structure** and is surrounded by a **membrane about 80 Å thick.**
- All eukaryotic cells have an ER. In animal cells, the ER usually forms more than half of the total membrane content.
- There are two types of endoplasmic reticulum:

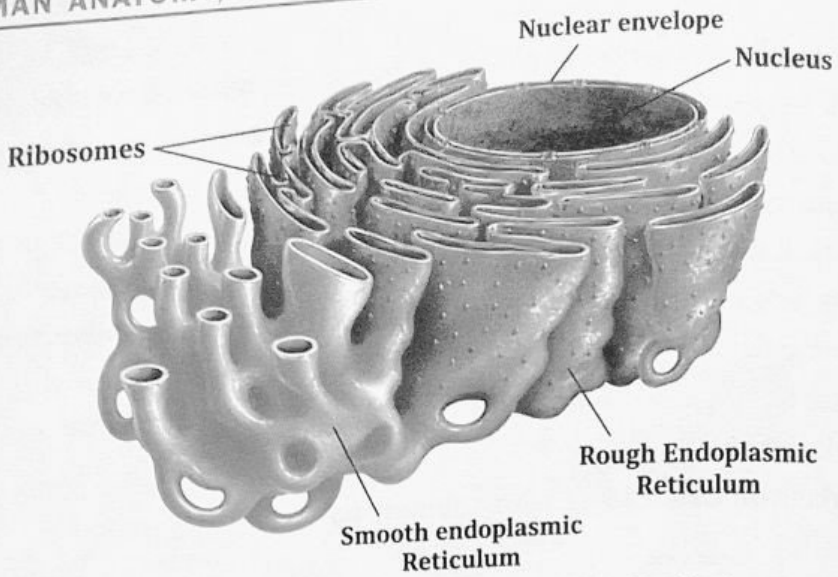


Fig.1.9: Structure of Endoplasmic Reticulum

✓ **Rough-Surfaced Endoplasmic Reticulum:**

- This type of reticulum contains osmophilic granules.
- **Ribosomes** are arranged in rows and **are attached to the membrane of the endoplasmic reticulum.**
- The rough appearance of the membrane is due to these granules.
- These **ribosomes** are the sites where proteins are synthesized.
- Some proteins, such as enzymes and hormones, may leave the cell and be used in other parts of the body.

✓ **Smooth-Surfaced Endoplasmic Reticulum:**

- There are no osmophilic granules in this type of endoplasmic reticulum.
- Ribosomes are present on the outer side of the membrane. Because of this, it appears smooth.
- The smooth endoplasmic reticulum carries out many metabolic processes.
- It produces lipids, phospholipids (for the plasma membrane), and steroids.
- These substances are secreted by cells of the testes, ovaries, and sebaceous glands.

✓ **Functions:**

- Protein synthesis takes place in the **rough endoplasmic reticulum**, which is found in large amounts in liver cells.
- Mammalian liver and gonad cells include the smooth endoplasmic reticulum, which is involved in making steroid hormones and may act as a detoxifying site in other cells.
- The sarcoplasmic reticulum in skeletal muscle plays a part in carrying impulses in the substance of muscle cells and is also involved in the binding of Ca^{2+} ions.

➢ **Golgi Apparatus (Golgi Complex)**

- The **Golgi apparatus** was discovered by **Camillo Golgi in 1898.**
- It is a complex structure made of interconnected **tubules, vesicles, and cisternae.**
- It is present in most eukaryotic cells and is usually located near the nucleus and endoplasmic reticulum (ER).
- The Golgi apparatus is a membrane-bound organelle responsible for modifying, sorting, and packaging proteins and lipids for transport.
- It exhibits structural polarity with a cis (forming) face and a trans (maturing) face.
- Under an electron microscope, its main structures include:

Unlock FREE B.Pharm 1st Semester Question Practice & e-PDFs - Download GDC CLASSES App Now

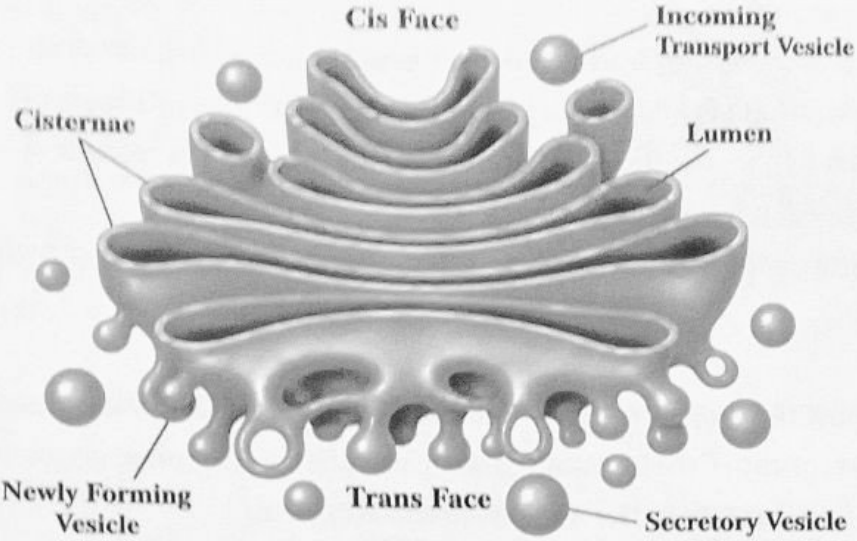


Fig.1.10: Structure of the Golgi Apparatus

✓ **Cisternae:**

- These are flattened sacs arranged in stacks. In longitudinal view, they appear like tubules.
- Each stack is called a dictyosome and usually contains 5–6 cisternae in animal cells.

✓ **Vesicles:**

- **Transitional vesicles:** Small membrane-bound vesicles formed from the ER. They move to the cis face of the Golgi and fuse to form new cisternae.
- **Secretory vesicles:** These vesicles become dense after filling with protein material. They release their contents outside the cell by exocytosis in response to signals.
- **Clathrin-coated vesicles:** These are small, spherical vesicles with a rough surface, found at the periphery of the Golgi. They are different from secretory vesicles and help in **transport of materials within the cell** (intracellular transport/pinocytosis).

✓ **Functions of Golgi Apparatus**

- It acts as a site for the **storage, processing, and packaging of cell products.**
- It modifies, concentrates, and transports substances received from the endoplasmic reticulum (ER).
- It helps in the formation of the plasma membrane, cell wall (in plants), and lysosomes.
- It **produces secretory granules and vacuoles** containing substances such as hormones, enzymes, and proteins.
- It synthesizes the polysaccharide part of glycoproteins.

➢ Lysosomes

- Lysosomes are small membrane-bound sacs (0.25–0.50 micron in size) **that contain digestive enzymes.**
- These enzymes break down large molecules like proteins, carbohydrates, lipids, and nucleic acids into smaller parts, which can then be used by the cell.
- Sometimes, **lysosomes can digest the cell's own contents**, so they are also called **"suicide bags."**

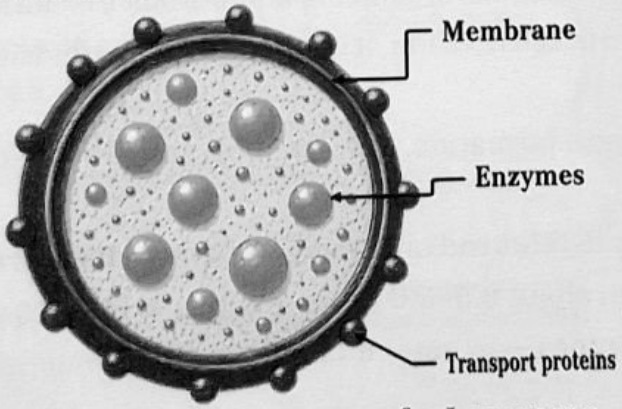


Fig.1.11: Structure of a Lysosome

✓ **Functions of Lysosomes**

- **Digestion:** They carry out intracellular digestion, so they are called the digestive system of the cell.
- **Cell necrosis:** When a cell is damaged, lysosomal enzymes are released and digest the cell parts.
- **Phagocytosis**: In phagocytosis, lysosomes join with phagosomes to form phagolysosomes and release enzymes to digest bacteria and other particles.
- **Autolysis:** After cell death, lysosomal enzymes break down the remaining cell parts, hence called suicide bags.

➢ **Centrosome**

- The centrosome is a specialized area of clear cytoplasm called the centrosphere.
- It contains two or more dark-staining particles called **centrioles,** which are usually arranged in pairs (diplosome) and lie close to the nucleus in a resting cell.
- It also contains a complex of proteins that help in the formation of microtubules.

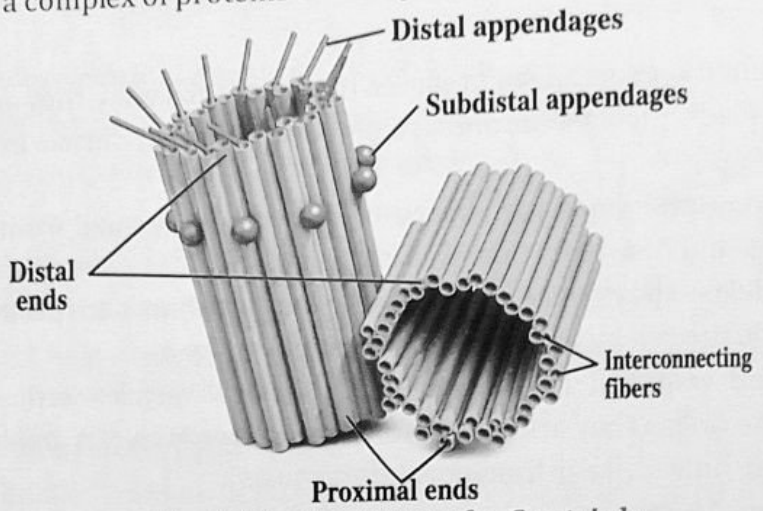


Fig.1.12: Structure of a Centriole

✓ **Structure of centrioles**

- Electron microscopy shows that a **centriole is a hollow cylindrical structure** about 3–5 μm long.
- Its wall is made of nine sets of microtubules arranged lengthwise, and each set has three subunits (triplets).
- Centrioles play an important role in **forming the spindle during mitosis.**
- From each centriole, microtubules spread out in a star-like pattern called an aster.
- These asters grow and move to opposite poles of the cell.
- Their fibers meet at the center (equator) of the cell and form the spindle.
- After cell division, these microtubules disappear.

✓ **Functions**

- They act as the main centers for organizing microtubules in animal cells.
- The centrosome helps in cell division. It duplicates before division, and the two centrosomes move to opposite ends of the cell.
- Centrioles help in spindle formation, cell signaling, and provide support to the cell.

➢ **Mitochondria**

- Mitochondria are small, solid, rod or **sausage-shaped structures** present in the cytoplasm.
- Their size ranges from about **0.5–5.0 μm** in length and **0.5–1.0 μm** in diameter.
- They have two membranes—an outer membrane and an inner membrane—and a gel-like inner part called the matrix.

Unlock FREE B.Pharm 1st Semester Question Practice & e-PDFs - Download GDC CLASSES App Now

- The outer membrane is smooth and contains special proteins called porins, which allow ions and molecules like **ADP and ATP** to pass easily.
- The inner membrane is folded to form cristae.
- These folds increase surface area and help in energy production.
- The inner membrane contains enzymes of the electron transport chain, which are responsible for ATP formation.
- The number and size of mitochondria depend on the energy needs of the cell.
- They are more numerous in active cells and are found in high numbers in organs like the liver, kidney, and heart.
- Some mitochondrial disorders include **Alpers disease, Barth syndrome, and Kearns–Sayre syndrome**.

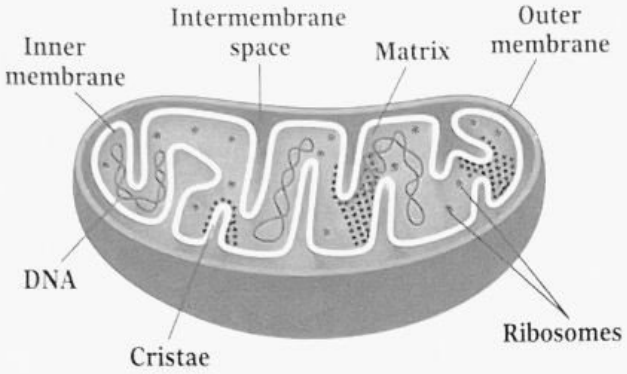


Fig.1.13: Structure of a Mitochondrion

✓ **Functions of Mitochondria**

- Performs oxidative phosphorylation to **produce ATP, the main energy source of the cell.**
- Krebs cycle takes place inside mitochondria.
- Contain respiratory enzymes (flavoproteins, cytochromes).
- Supply about 95% of the cell's energy. Hence, called the **"Powerhouse of the cell".**
- Mitochondria have a small amount of DNA, which helps in their replication.
- This DNA controls the division of mitochondria inside the cell.
- They play an important role in **apoptosis (programmed cell death).**

➢ Ribosomes

- Ribosomes are small, non-membrane-bound organelles composed of rRNA and proteins, and their main function is protein synthesis.
- Each ribosome consists of **two subunits (large and small)**, which are formed separately in the **nucleolus** and unite in the cytoplasm.
- Ribosomes attached to the nuclear membrane and rough endoplasmic reticulum (RER) synthesize proteins for organelles, the plasma membrane, and secretion outside the cell.
- **Free ribosomes** float in the cytoplasm and produce proteins used within the cytosol.
- Ribosomes are also present in **mitochondria**, where they synthesize mitochondrial proteins.

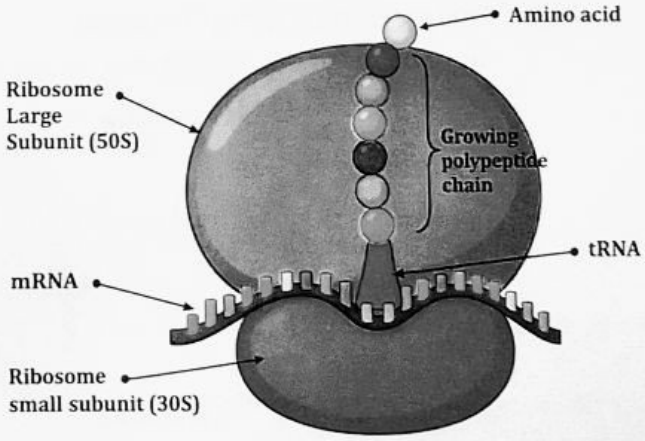


Fig .1.14: Structure of a Ribosome

✓ **Functions of Ribosomes**

- **Translation of mRNA:** Ribosomes read the genetic code carried by messenger RNA (mRNA).
- **Protein Synthesis:** They join amino acids together in the correct sequence to form proteins.
- **Binding of tRNA:** Ribosomes provide sites for transfer RNA (tRNA) to attach and deliver amino acids.
- **Catalyzing Peptide Bond Formation:** The large ribosomal subunit acts as an enzyme (ribozyme) that forms peptide bonds between amino acids.
- **Ensuring Accuracy of Protein Production:** They help match the correct tRNA anticodon with the mRNA codon, ensuring proper protein formation.

➢ **Peroxisomes**

- These are **small membrane-bound organelles**, also called microbodies, that contain oxidase and catalase enzymes.
- They break down fatty acids, amino acids, and toxic substances such as alcohol, and are especially abundant in the liver.
- **They produce and detoxify hydrogen peroxide (H_2O_2),** protecting cells from oxidative damage and free radicals.
- Peroxisomes can self-replicate by growing and dividing or by forming from cellular components.

✓ **Functions of Peroxisomes**

- β-oxidation of fats.
- α-oxidation of phytanic acid.
- Breakdown of hydrogen peroxide.
- Formation of bile acids.
- Formation of plasmalogens.

➢ **Inclusions**

These are non-living substances in the cell that are not surrounded by a membrane, such as fat droplets, glycogen granules, and pigment granules.

❖ **Nucleus**

- The nucleus, which is located in the **middle of the cell, is typically a spherical body.**
- It varies in size, shape, quantity, and location.
- All human cells have a nucleus, **except mature red blood cells.**
- Some cells, like skeletal muscle cells, have more than one nucleus.
- The nucleus is the largest organelle and is surrounded by a nuclear envelope.
- This envelope is similar to the cell membrane and has tiny pores that allow some large substances to move between the nucleus and the cytoplasm.

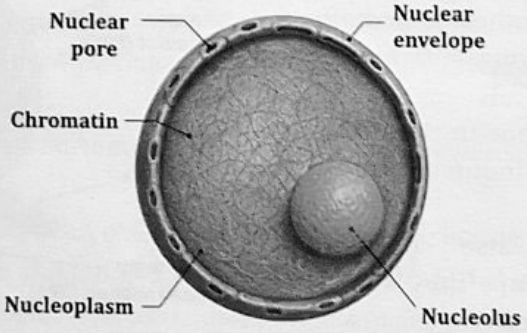


Fig.1.15: Structure of the Nucleus

Unlock FREE B.Pharm 1st Semester Question Practice & e-PDFs - Download GDC CLASSES App Now

➢ **Nucleolus**
- Inside the nucleus, there is usually one, or sometimes 2–5 small bodies called nucleolus (nucleoli).
- The nucleolus is made of an uneven network or rows of tiny granules called nucleonema, seen under an electron microscope.
- During cell division, the nucleolus disappears.

➢ **Chromatin**
- Chromatin contains many genes that regulate the cell's heredity as well as the reassembly of distinct chromatins to create a chromosome.
- Chromatin is visible under a microscope in a non-dividing cell (interphase) as dark, dispersed sections inside the nucleus.
- There are two kinds of chromatin: **euchromatin and heterochromatin.**
- Euchromatin is loosely coiled and active, whereas heterochromatin is firmly coiled and inert.
- Euchromatin contributes to the production of messenger RNA (mRNA).

➢ **Chromosomes**
- Chromosomes appear as separate bodies during cell division.
- Their main component is **DNA and histone proteins.**
- A gene is a part of DNA present on a chromosome and **carries hereditary information** for a specific trait.
- Each chromosome has coiled threads called chromonema and small particles called chromomeres, which contain DNA.
- Chromosomes are classified based on the position of the centromere:
  - ✓ Near the end → **Acrocentric**
  - ✓ In the center → **Metacentric**
  - ✓ Slightly off-center → **Submetacentric**

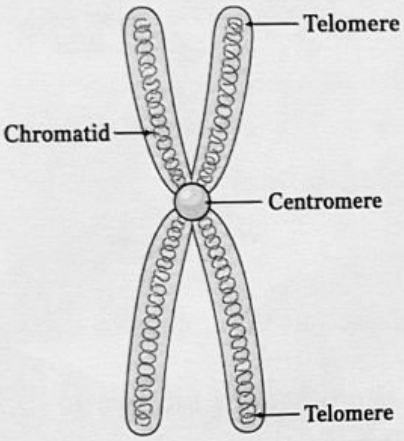


Fig.1.16: Eleven Major Organ Systems of the Human Body

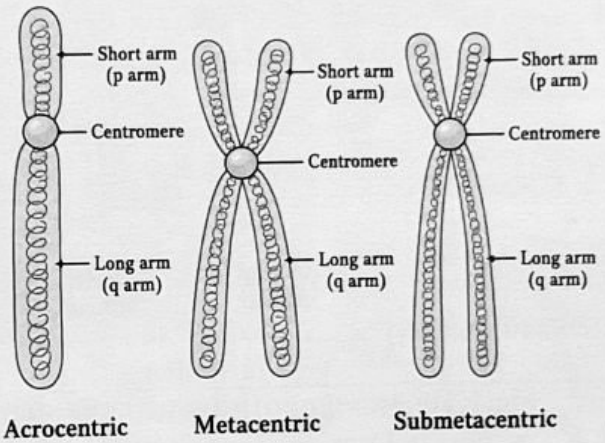


Fig.1.17: Types of Chromosomes Based on the Position of the Centromere

➢ **Autosomes and Sex Chromosomes**
- Humans have **46 chromosomes** in each cell, arranged in **23 pairs.**
- Out of these, **22 pairs are autosomes,** which control general body traits.
- The remaining **1 pair are sex chromosomes**, which decide gender.
- In females, the sex chromosomes are **XX**, and in males, they are **XY**. One X chromosome in females may appear as a Barr body under a microscope.
- In reproductive cells (gametes), females have **22 autosomes and one X chromosome,** while males have **22 autosomes and either an X or Y chromosome.**
- Chromosomes carry hereditary information and pass traits from one generation to the next.

✓ **Functions**

- **Protein synthesis**: The nucleus makes messenger RNA (mRNA), which carries genetic information. This mRNA comes out of the nucleus through pores and goes to ribosomes in the cytoplasm, where proteins are made. It attaches to ribosomes and helps in protein formation.
- **Carrier of heredity:** The nucleus stores genetic information and passes it from parents to offspring.

1.2.2. Transport Across Cell Membrane

- The cell membrane helps in the transport of materials in and out of the cell.
- It is selectively permeable, meaning it allows only lipophilic substances to pass across, while the hydrophilic substances are transported through specialized proteins.
- So, the cell membrane works as a selective barrier.
- There are mainly two processes by which substances move across it.
  - ✓ **Passive Transport**
  - ✓ **Active Transport**

❑ Passive Transport

Substances can cross the semipermeable plasma and organelle membranes, and move down the concentration gradient (downhill) without using energy.

❖ **Diffusion:** Lipid-soluble substances such as oxygen and carbon dioxide, along with small molecules, pass directly across the cell membrane.

❖ **Facilitated diffusion:** Water-soluble substances such as sodium and calcium move across the membrane through protein channels or carrier proteins.

❖ **Osmosis:** The movement of a solvent across a semipermeable membrane from a region of lower solute concentration to a region of higher solute concentration. In cells, the solvent is usually water. Osmosis helps in nutrient absorption and in removing waste from the blood.

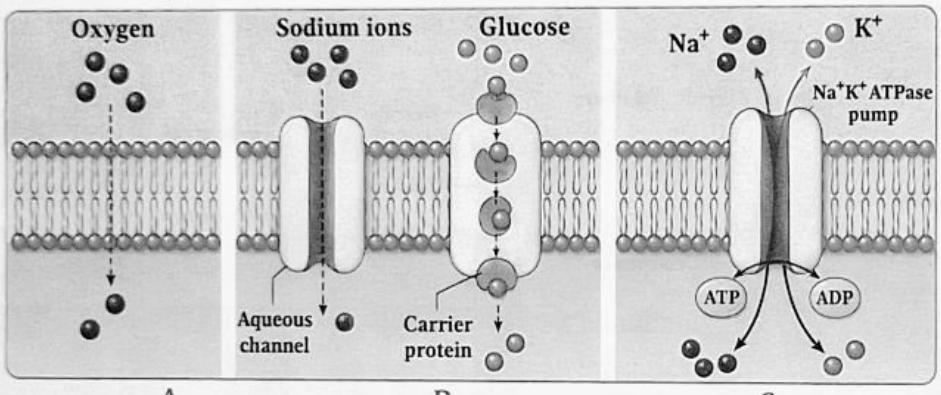


Fig.1.18: Movement of molecules across the plasma membrane.
A. Simple diffusion; B. Facilitated diffusion; C. Active transport

❑ Active Transport

- Active transport is the movement of substances from a lower to a higher concentration using energy in the form of ATP.
- It involves specific carrier proteins in the cell membrane, and the transport rate depends on the number of available carriers.

❖ **Endocytosis**

- Endocytosis is an active transport process in which eukaryotic cells take in external substances such as nutrients, fluids, and pathogens—by enclosing them in vesicles formed from the plasma membrane.

  It is of two types:

  - ✓ **Phagocytosis (cell eating):** Uptake of large particles like bacteria or cell debris by specialized cells.

Unlock FREE B.Pharm 1st Semester Question Practice & e-PDFs - Download GDC CLASSES App Now

✓ **Pinocytosis (cell drinking):** Uptake of surrounding fluid and dissolved substances through small vesicles.

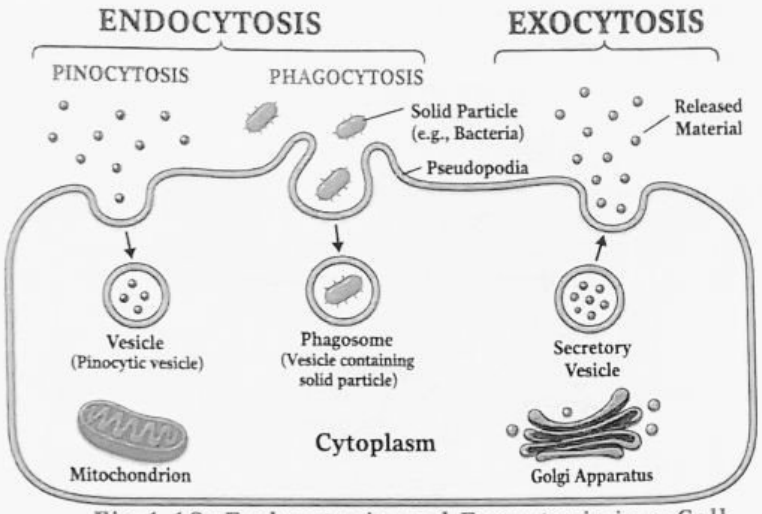


Fig.1.19: Endocytosis and Exocytosis in a Cell

❖ **Exocytosis**

- Exocytosis is the process by which cells expel waste or undigested materials and release substances like enzymes and hormones outside the cell.

❖ **Sodium-Potassium Pump**

- The sodium-potassium pump **uses ATP to maintain an electrical gradient** across the cell membrane, which is essential for nerve and muscle function.
- It keeps **potassium levels higher inside the cell and sodium levels higher outside** by moving potassium in and pumping sodium out in opposite directions.

Table 1.7: Difference between Active and Passive Transport

PASSIVE TRANSPORT	ACTIVE TRANSPORT
Movement of substances occurs without energy expenditure.	Movement of substances requires energy (ATP).
Substances move along the concentration gradient (high to low concentration).	Substances move against the concentration gradient (low to high concentration).
Does not require carrier proteins in simple diffusion.	Usually requires carrier proteins or pumps.
It is a rapid process.	It is a comparatively slower process.
Includes diffusion, osmosis, and facilitated diffusion.	Includes primary and secondary active transport.
Cannot accumulate substances beyond equilibrium.	Can accumulate substances inside the cell against equilibrium.
Example: Movement of oxygen and carbon dioxide across the membrane.	Example: Sodium-potassium pump (Na^+/K^+ pump).

1.2.3. Cell Cycle and Cell Division

❑ **The Cell Cycle**

- The cell cycle is a regulated sequence of events in which a eukaryotic **cell copies its DNA,** produces other cell components, and **divides to form two identical daughter cells.**
- The **period between two cell divisions** is known as the cell cycle.
- The duration of the cell cycle varies among different cell types; in rapidly dividing human cells, it is typically about **24 hours.**
- It ensures accurate replication and equal distribution of DNA.
- The cycle has two main stages: